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Studierichting: Informatica
Oefeningenreeks 4A nr. 44.2

$$\begin{aligned}\text{opgave: } & \int \frac{3x^2 + x + 4}{x^3 + x^2 + x + 1} dx \\ \Rightarrow & \frac{3x^2 + x + 4}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1} \\ \Rightarrow A = & \left. \frac{3x^2 + x + 4}{x^2 + 1} \right|_{x=-1} = 3 \\ \Rightarrow Bi + C = & \left. \frac{3x^2 + x + 4}{x + 1} \right|_{x=i} = \frac{i+1}{i+1} = 1 \\ \frac{3x^2 + x + 4}{(x+1)(x^2+1)} = & \frac{3}{x+1} + \frac{1}{x^2+1} \\ I = & \int \frac{3}{x+1} dx + \int \frac{1}{x^2+1} dx \\ = & 3 \ln|x+1| + \text{Bgtan}(x) + c\end{aligned}$$

References